Dylan Trembly

Instructions:
- Complete and submit the final file by the deadline.
- No make-up exams will be provided.
- A total of 70 points.

Q1 (3 points): The scatter plot below shows the relationship between GDP per capita (in USD) and life satisfaction for various countries.Based on this plot, which of the following statements best reflects the insights that can be gathered and how they relate to a machine learning model that could be built using this data?

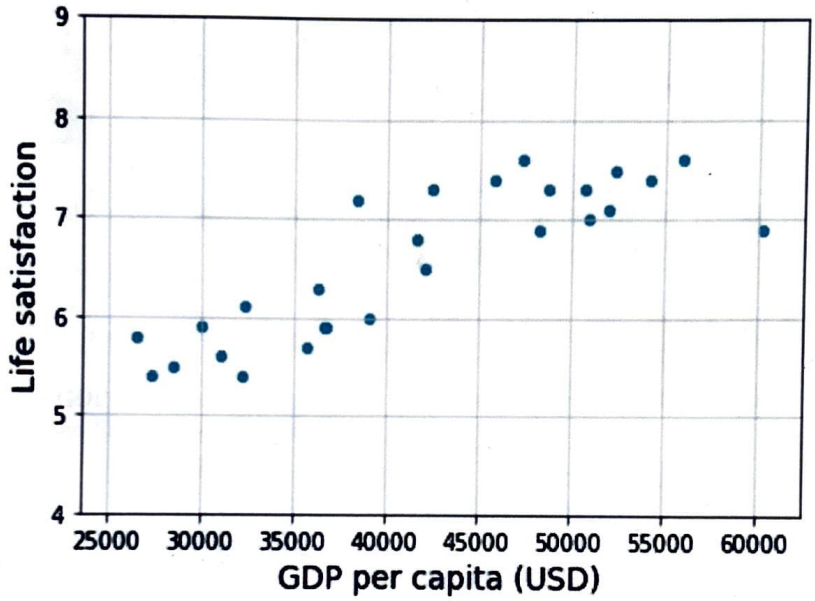

Figure 1: Scatter plot of GDP per capita vs. Life satisfaction

A) The plot indicates a strong negative correlation between GDP per capita and life satisfaction, suggesting that as GDP increases, life satisfaction decreases.

B) The scatter plot suggests a positive correlation between GDP per capita and life satisfaction, which could indicate that a linear regression model might be a suitable approach for predicting life satisfaction based on GDP per capita.

C) The distribution of points indicates no clear relationship between GDP per capita and life satisfaction, suggesting that a machine learning model would not be useful for predicting life satisfaction.

D) The data points show a perfect linear relationship, implying that any machine learning model would achieve 100% accuracy when predicting life satisfaction.

Q2 (5 points):The graph below shows the relationship between GDP per capita (in USD) and life satisfaction, along with a fitted linear regression line. The equation for the line is provided on the graph, where $\theta_0 = 3.75$ and $\theta_1 = 6.78 \times 10^{-5}$.

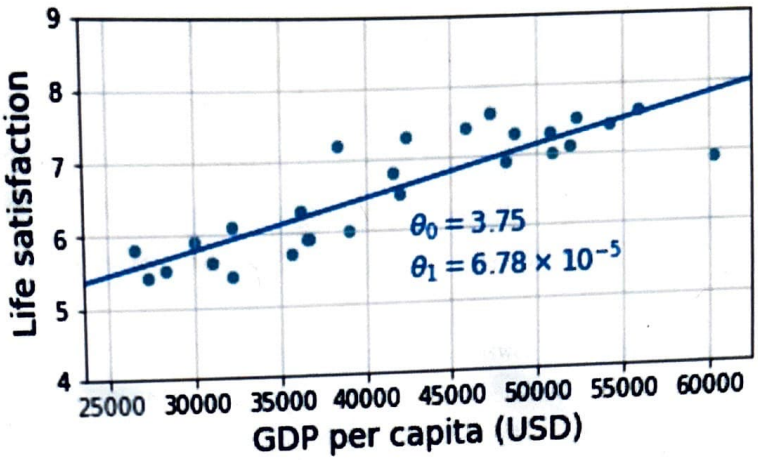

Figure 2: Scatter plot with regression line for GDP per capita vs. Life satisfaction

Given that the GDP per capita of Country X is \$37,655.18, use the information from the graph to calculate the predicted life satisfaction score for Country X.

$y = 3.75 + 6.78 \times 10^{-5}(x)$

$x = 37655.18$

$y = 3.75 + 6.78 \times 10^{-5}(37655.18)$

$y = 6.303$

predicted life satisfaction:

6.303

Q3 (2 points): Given the following Python code snippet below, assume that `lin_reg` is a trained linear regression model and **housing** is a DataFrame with data used for predictions. What will be the output of this code?

```
housing_predictions = lin_reg.predict(housing)
housing_predictions[:5].round(-2)
```

should be ":10"? Typo?

A) The first 10 predicted values, rounded to the nearest hundred.

B) The last 10 predicted values, rounded to the nearest ten.

C) The entire set of predictions, rounded to the nearest hundred.

D) The first 10 actual values from **housing**, rounded to the nearest hundred.

Q4 (6 points): Given the following Python code snippet, answer the following questions:

```
from sklearn.metrics import mean_squared_error
lin_rmse = mean_squared_error(housing_labels, housing_predictions, squared=False)
lin_rmse
Output: 68687.89176589991
```

1. What does the `mean_squared_error` function do when `squared=False`?

   A) It calculates the mean absolute error (MAE).

   B) It calculates the mean squared error (MSE) without squaring the residuals.

   C) It calculates the root mean squared error (RMSE).

   D) It calculates the variance of the predictions.

2. Based on the output `68687.89176589991`, what can be interpreted about the RMSE value in the context of the housing data?

   A) On average, the predicted housing prices differ from the actual prices by approximately $68,688.

   B) The model is perfect, with zero prediction error.

   C) The variance in housing prices is $68,688.

   D) The model underestimates prices by $68,688.

3. If the average house price in the dataset is around $250,000, what does an RMSE value of `68687.89176589991` suggest about the model's performance?

   A) The model's predictions are very close to the actual prices, showing high accuracy.

   B) The model's prediction error is relatively high, indicating that the model's predictions, on average, deviate significantly from the true house prices.

   C) The model's performance is nearly perfect, as the RMSE value is much smaller than the average price.

   D) The model's predictions are consistent but show a small bias.

Q5 (2 points): The results of predicting housing prices using a trained model indicate that the test RMSE is approximately \$41,424. Additionally, a 95% confidence interval for the test RMSE has been computed as [\$39,275, \$43,467]. What does this 95% confidence interval imply about the model's performance when predicting housing prices?

A) The model's predictions are guaranteed to be within the range of \$39,275 and \$43,467 for every test instance.

B) There is a 95% probability that the true RMSE of the model on the test set lies between \$39,275 and \$43,467.

C) The average error for each prediction will always fall between \$39,275 and \$43,467.

D) The interval suggests that the prediction errors will vary between \$39,275 and \$43,467 for most data points.

Q6 (2 points): The RMSE for the training set when predicting housing prices is approximately \$68,688, while the RMSE for the test set is approximately \$41,424. Which of the following interpretations or insights is most appropriate based on these results?

A) The model performs significantly better on the training set than on the test set, indicating underfitting.

B) The lower RMSE on the test set compared to the training set suggests the model might be overfitting the training data.

C) The higher RMSE on the training set indicates that the model has good generalization ability to new, unseen data.

D) The results suggest that the model has poor performance on both the training and test sets, and further optimization is needed.

Q7 (5 points): For a model with an R^2 value of 0.85 and 100 observations, the adjusted R^2 is requested. Assume the model includes 3 predictors. Calculate the adjusted R^2.

$$\text{adj } R^2 = 1 - \frac{(1-R^2)(n-1)}{n-p-1} = 1 - \frac{(1-.85)(100-1)}{100-3-1} = .845$$

adj $R^2 = .845$

Q8 (10 points): Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce the estimated coefficients:

$$\hat{\beta}_0 = -6, \quad \hat{\beta}_1 = 0.05, \quad \hat{\beta}_2 = 1.$$

- Estimate the probability that a student who studies for 40 hours and has an undergrad GPA of 3.5 gets an A in the class.

$\ln\left(\frac{p}{1-p}\right) = B_0 + B_1X_1 + B_2X_2$ $\quad z = -6 + .05(40) + 3.5 = -.5$

$e^{-.5} = \frac{p}{1-p} \rightarrow \frac{.606}{1+.606} = p, \; p = .378$

- Calculate and interpret the odds ratio for a one-unit increase in X_1 (hours studied). What does this imply in the context of the problem regarding the likelihood of receiving an A?

$z = -6 + .05(41) + 3.5 = -.45$

$e^{-.45} = .638, \; p = .389$

$\frac{.638}{.606} = 1.051$

5.1% odds increase to get an A if the student studies for 41 hours instead of 40 (1 hour increase).

Q9 (15 points): Suppose you are given the following logistic regression equation:

$$\hat{p} = \frac{\exp(-3 + 0.02x_1 + 0.5x_2)}{1 + \exp(-3 + 0.02x_1 + 0.5x_2)}$$

where x_1 is hours worked per week and x_2 is whether the employee has completed safety training (0 = No, 1 = Yes).

- Calculate the probability that an employee who works 40 hours/week and has completed training will exhibit a safe behavior.
- Interpret the coefficient of x_2 in terms of odds ratio.

$$p = \frac{e^{-3+.02(40)+.5}}{1+e^{-3+.02(40)+.5}} = \boxed{.1545}$$

odds ratio for x_2

$e^{.5} = 1.649$

completing training increases safe behavior odds by 64.9%

Q10 (10 points): The following multiple linear regression model was fit:

$$\hat{y} = 150 + 2.5x_1 - 1.8x_2 + 0.75x_3$$

where y = productivity score, x_1 = training hours, x_2 = absentee days, x_3 = years of experience.

- Predict the productivity score for an employee with $x_1 = 40$, $x_2 = 3$, $x_3 = 5$.
- Interpret the effect of each coefficient.

$y = 150 + 2.5(40) - 1.8(3) + .75(5) = 248.35$

base intercept = 150

each hour trained = +2.5 productivity

each absent day = −1.8 productivity

each year of exp = +.75 productivity

training hours have the biggest impact (excluding intercept)

Q11 (10 points): Suppose you have a logistic regression output for a predictor X:

$$\hat{\beta} = 0.8, \quad \mathrm{SE}(\hat{\beta}) = 0.25$$

Using the model output, perform a Wald test to assess if the predictor X is statistically significant at the 0.05 level. Outline the null and alternative hypotheses, compute the test statistic, and determine if the null hypothesis can be rejected.

$$t = \frac{.8}{.25} = 3.2$$

Statistical significance of .05 gives our z-number to be ±1.96 (no n for t-dist. given).

$H_0: \hat{\beta} = 0$, $H_a: \hat{\beta} \neq 0$

$3.2 > |\pm 1.96|$, so we can reject the H_0.

X is statistically significant